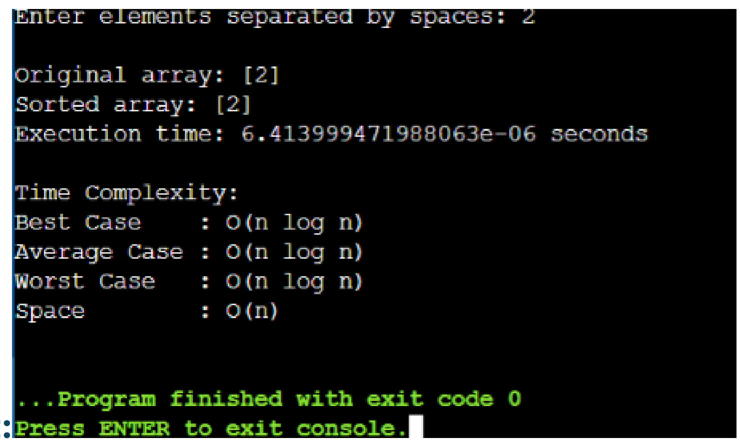
Merge Sort Definition

Merge Sort is a **divide-and-conquer sorting algorithm**. It divides an array into smaller sub-arrays, sorts them recursively, and then merges the sorted sub-arrays to produce a final sorted array.

Algorithm of Merge Sort

1. Start with an unsorted array.
2. Find the middle position of the array.
3. Divide the array into two halves: **left half** and **right half**.
4. Recursively apply Merge Sort to both halves.
5. Compare the elements of the two sorted halves.
6. Merge them in ascending order.
7. Continue until the complete array is sorted.

Output:



```
Enter elements separated by spaces: 2

Original array: [2]
Sorted array: [2]
Execution time: 6.4139994719880630e-06 seconds

Time Complexity:
Best Case     : O(n log n)
Average Case  : O(n log n)
Worst Case    : O(n log n)
Space         : O(n)


...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

Case	Time Complexity
Best Case	**O(n log n)**
Average Case	**O(n log n)**
Worst Case	**O(n log n)**

Space Complexity

O(n) — Merge Sort requires additional space for merging the sub-arrays.

Short Answer for Exam

Merge Sort is a divide-and-conquer sorting algorithm that divides an array into two halves, recursively sorts each half, and merges the sorted halves. Its best, average, and worst-case time complexity is **O(n log n)**, while its space complexity is **O(n)**.